CO1 – ASSESSMENT NO 2

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12. Design
a machine learning model for spam email classification using labeled data.

(a) Explain the choice of algorithm
(e.g., Naïve Bayes, SVM, or Logistic Regression).

(b) Describe the preprocessing steps
such as tokenization, stop-word removal, and vectorization.

(c) Discuss evaluation metrics such as
accuracy, precision, recall, and F1-score.

(A) CHOICE OF ALGORITHM

Several machine learning algorithms can be used for spam classification.

1. Naïve Bayes (Most Common)

Reason for choosing:

- Works well with text data.
- Fast to train and predict.
- Handles large datasets efficiently.
- Based on probability using Bayes' theorem.
- Suitable for high-dimensional features like words in emails.

Working:

- Calculates the probability that an email belongs to the spam class.
- Chooses the class with the highest probability.

Advantages

- Very fast
- High accuracy for text classification

- Requires less training data
- Easy to implement

Disadvantages

- Assumes features are independent, which is not always true.

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2. Support Vector Machine (SVM)

Reason for choosing:

- Creates the best boundary (hyperplane) between spam and ham emails.
- Performs well on high-dimensional text data.

Advantages

- High classification accuracy
- Effective with complex datasets

Disadvantages

- Slower on very large datasets
- Requires more computational resources

(B) PREPROCESSING STEPS

Remove unnecessary characters such as:

- Punctuation
- Numbers
- HTML tags
- Special symbols

Example:

Original

"Congratulations!!! You won $5000."

After Cleaning

"congratulations you won"

Step 3: Convert to Lowercase

Convert all text to lowercase.

Example:

"FREE MONEY"

↓

"free money"

Step 4: Tokenization

Split the sentence into individual words.

Example:

"Win free money now"

↓

["win", "free", "money", "now"]

Step 5: Stop-word Removal

Remove commonly occurring words that do not add much meaning.

Examples:

- is
- the
- am
- are
- and
- of
- in

Example:

"This is a free offer"

↓

["free", "offer"]

Step 6: Stemming or Lemmatization

Convert words to their root form.

Examples:

Running → Run

Playing → Play

Offers → Offer

Step 7: Vectorization

Machine learning algorithms cannot understand text directly.

Convert words into numbers using:

- **Bag of Words (BoW):** Counts how many times each word appears.
- **TF-IDF (Term Frequency–Inverse Document Frequency):** Gives higher importance to informative words and lower importance to common words.

(C) EVALUATION METRICS

After training, evaluate the model using the following metrics.

1. Accuracy

Measures the percentage of correctly classified emails.

Formula:

Accuracy = (Correct Predictions / Total Predictions) × 100

Example

Correct predictions = 950

Total emails = 1000

Accuracy = 95%

Interpretation: The model correctly classifies 95% of emails.

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2. Precision

Measures how many emails predicted as spam are actually spam.

Formula:

Precision = TP / (TP + FP)

Where:

- **TP (True Positives):** Spam correctly identified as spam.
- **FP (False Positives):** Normal emails incorrectly marked as spam.

Importance: High precision reduces the chance of legitimate emails being marked as spam.

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3. Recall

Measures how many actual spam emails are correctly detected.

Formula:

Recall = TP / (TP + FN)

Where:

- **FN (False Negatives):** Spam emails incorrectly classified as normal.

Importance: High recall ensures fewer spam emails reach the user's inbox.

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4. F1-Score

The harmonic mean of Precision and Recall.

Formula:

F1-score = 2 × (Precision × Recall) / (Precision + Recall)

Importance: Provides a balanced measure when both precision and recall are important, especially for imbalanced datasets.